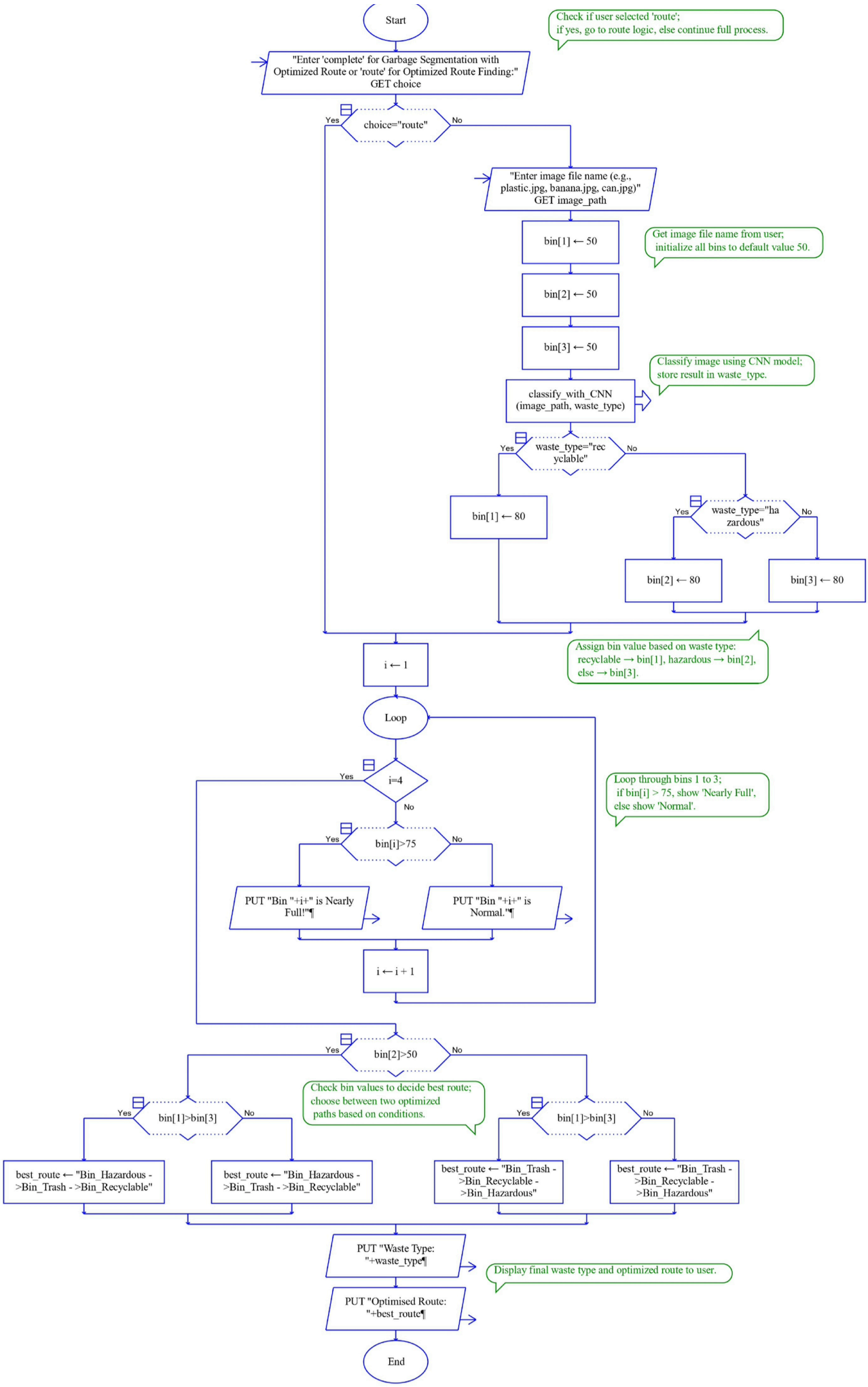
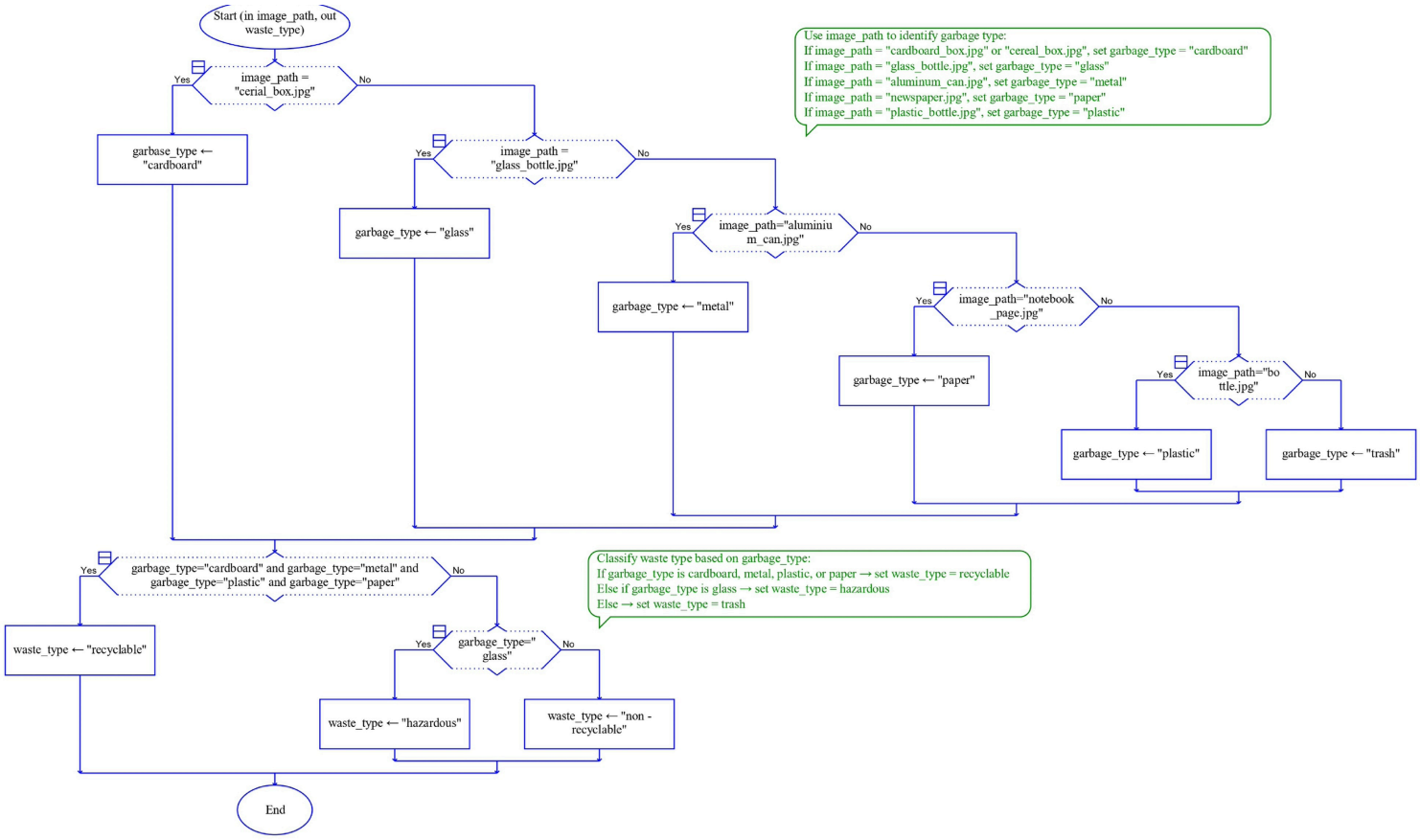

Start
Check if user selected 'route';
if yes, go to route logic, else continue full process.
"Enter 'complete' for Garbage Segmentation with Optimized Route or 'route' for Optimized Route Finding:" GET choice
choice="route"
Yes
No
"Enter image file name (e.g., plastic.jpg, banana.jpg, can.jpg)" GET image_path
bin[1] ← 50
Get image file name from user;
initialize all bins to default value 50.
bin[2] ← 50
bin[3] ← 50
Classify image using CNN model;
store result in waste_type.
classify_with_CNN (image_path, waste_type)
waste_type="recyclable"
bin[1] ← 80
waste_type="hazardous"
bin[2] ← 80
bin[3] ← 80
Assign bin value based on waste type:
recyclable → bin[1], hazardous → bin[2],
else → bin[3].
i ← 1
Loop
i=4
Loop through bins 1 to 3;
if bin[i] > 75, show 'Nearly Full',
else show 'Normal'.
bin[i]>75
PUT "Bin "+i+" is Nearly Full!"¶
PUT "Bin "+i+" is Normal."¶
i ← i + 1
bin[2]>50
Check bin values to decide best route;
choose between two optimized paths based on conditions.
bin[1]>bin[3]
best_route ← "Bin_Hazardous ->Bin_Trash - >Bin_Recyclable"
best_route ← "Bin_Hazardous ->Bin_Trash - >Bin_Recyclable"
best_route ← "Bin_Trash ->Bin_Recyclable - >Bin_Hazardous"
best_route ← "Bin_Trash ->Bin_Recyclable - >Bin_Hazardous"
PUT "Waste Type: "+waste_type¶
Display final waste type and optimized route to user.
PUT "Optimised Route: "+best_route¶
End


Start (in image_path, out waste_type)
image_path = "cerial_box.jpg"
Yes
No
garbase_type ← "cardboard"
image_path = "glass_bottle.jpg"
Yes
No
garbage_type ← "glass"
image_path="aluminium_can.jpg"
Yes
No
garbage_type ← "metal"
image_path="notebook_page.jpg"
Yes
No
garbage_type ← "paper"
image_path="bottle.jpg"
Yes
No
garbage_type ← "plastic"
garbage_type ← "trash"
Use image_path to identify garbage type:
If image_path = "cardboard_box.jpg" or "cereal_box.jpg", set garbage_type = "cardboard"
If image_path = "glass_bottle.jpg", set garbage_type = "glass"
If image_path = "aluminum_can.jpg", set garbage_type = "metal"
If image_path = "newspaper.jpg", set garbage_type = "paper"
If image_path = "plastic_bottle.jpg", set garbage_type = "plastic"
garbage_type="cardboard" and garbage_type="metal" and garbage_type="plastic" and garbage_type="paper"
Yes
No
waste_type ← "recyclable"
garbage_type="glass"
Yes
No
waste_type ← "hazardous"
waste_type ← "non-recyclable"
Classify waste type based on garbage_type:
If garbage_type is cardboard, metal, plastic, or paper → set waste_type = recyclable
Else if garbage_type is glass → set waste_type = hazardous
Else → set waste_type = trash
End